

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Based on: arxiv.org/abs/2504.04715

The Model Substitution Problem

Economic Incentives Drive Covert Swaps

Hypothesis Testing Framework:

- H_0 (Null): $P_{\text{spec}}(y|x)$ - Genuine model served
- H_1 (Alt): $P_{\text{actual}}(y|x)$ - Substitute model served

Smaller Models

e.g., 8B instead of 70B
Massive compute savings

Quantization

FP8/INT8 instead of FP16
Reduces memory bandwidth

Fine-Tuned

Cheaper variants updated
to hide weaknesses

Different Family

Completely different architecture
disguised as the target

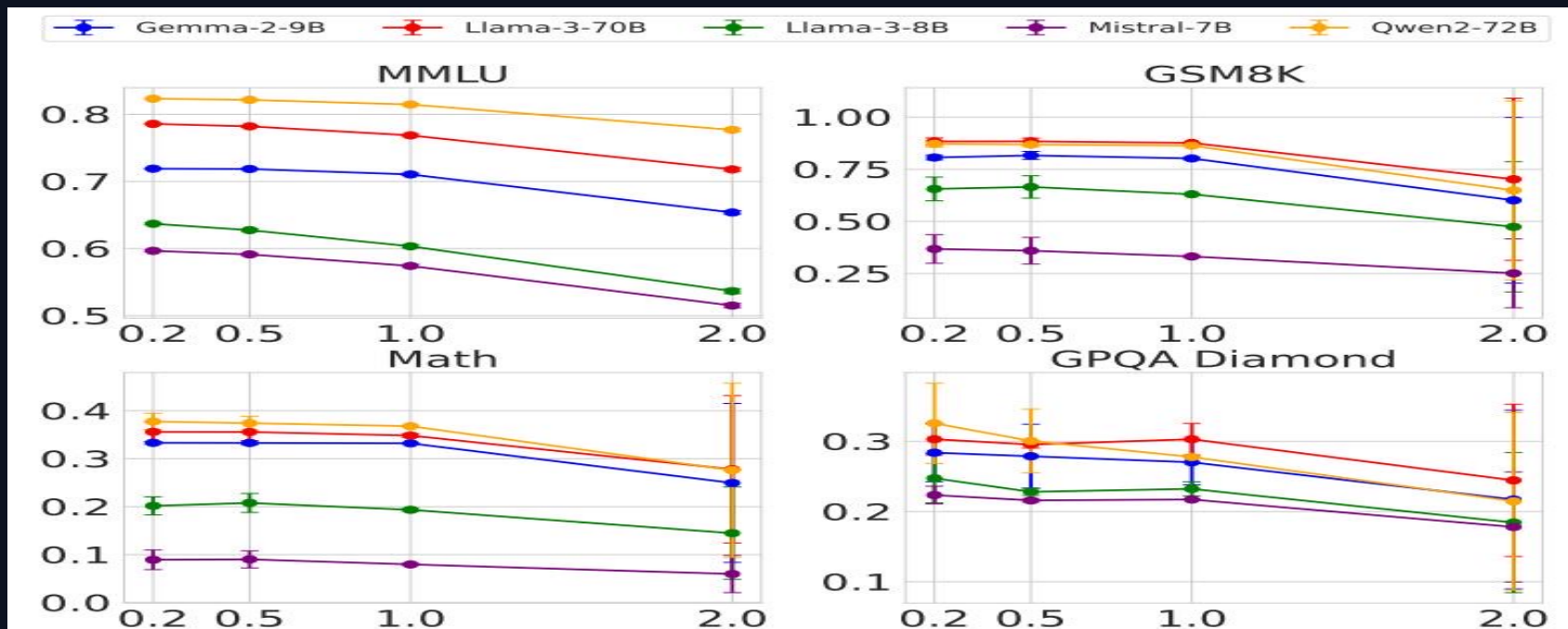
FP8 Quantization Preserves Most Accuracy

Making Substitution Hard to Detect (Scores often within error bars)

Model / Variant	MMLU	GSM8K	MATH	GPQA Dia.
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Higher Temperature Amplifies Gaps & Noise

At T=2.0, performance degrades globally, complicating statistical detection.



Log Probability Fingerprinting Fails

Inference nondeterminism across frameworks and hardware defeats metadata checks.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Randomized Routing is Query-Prohibitive

Attackers mix substitute responses with probability p

The Mixed Distribution

$$P_{mix} = (1-p) * P_{spec} + p * P_{alt}$$

As $p \rightarrow 0$, $P_{mix} \approx P_{spec}$

Statistical distance shrinks, requiring exponentially more queries to detect.

Benchmark Evasion

Providers detect known benchmark prompts and route them to the genuine model.

Result: High benchmark scores, but low-quality substitute responses for regular user traffic.

Software-Only Auditing Methods Fail

Why current approaches are fundamentally unreliable

Method Type	Core Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Fails against subtle substitutions (e.g., FP8)	Requires impractically large query budgets; defeated by small mixing ratios.
Log-Probability Fingerprinting	Production-level inference nondeterminism	Same model yields different log-probs across frameworks (vLLM) & hardware.
Adversarial Countermeasures	Vulnerable to prompt classification	Benchmark evasion makes standard query-based auditing useless.

Trusted Execution Environments (TEEs)

Provable Cryptographic Model Integrity

Hardware-Secured Enclaves

- Provides cryptographic attestation of code and data
- Guarantees specific model weights are loaded
- Guarantees correct inference code is running

Practical Deployment

- e.g., NVIDIA confidential computing on H100 GPUs
- Modest performance overhead
- Deployable today (unlike ZKPs which are computationally prohibitive)

Key Takeaways

Close the Verification Gap with Hardware Attestation

- Model substitution is economically motivated and practically undetectable via software.
- FP8 quantization barely affects benchmarks, providing an easy covert swap.
- Inference nondeterminism kills log-prob fingerprinting.
- Randomized substitution and benchmark evasion defeat statistical tests.
- TEEs are the only currently viable path to verifiable LLM APIs.

CALL TO ACTION: Demand TEE attestation for high-stakes API inference.